

# Charles Hong

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## EDUCATION

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| <b>University of California, Berkeley</b><br><i>Ph.D., Electrical Engineering and Computer Science</i> | Ongoing |
| <b>University of California, Berkeley</b><br><i>M.S., Electrical Engineering and Computer Science</i>  | 2025    |
| <b>University of California, Berkeley</b><br><i>B.S., Electrical Engineering and Computer Science</i>  | 2021    |

## HONORS/AWARDS

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|-----------------------------------------------------------------------------------|------|
| <b>LAD Fellowship</b> – IEEE                                                      | 2025 |
| <b>Master's Scholarship in Integrated Systems</b> – Apple                         | 2021 |
| <b>High Honors at Graduation</b> – University of California, Berkeley             | 2021 |
| <b>Eta Kappa Nu Honor Society</b> – IEEE                                          | 2018 |
| <b>Edward Frank Kraft Award for Freshmen</b> – University of California, Berkeley | 2017 |

## CONFERENCE PAPERS

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| <b>hdl2v: A Code Translation Dataset for Enhanced LLM Verilog Generation</b> ( <a href="#">Paper</a> ) ( <a href="#">Dataset</a> )<br>Charles Hong, Brendan Roberts, Huijia An, Alex Um, Advay Ratan, Yakun Sophia Shao<br><i>7th ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD), September 2025</i>                                          |
| <b>DOSA: Differentiable Model-Based One-Loop Search for DNN Accelerators</b> ( <a href="#">Paper</a> ) ( <a href="#">Slides</a> ) ( <a href="#">GitHub</a> )<br>Charles Hong, Qijing Huang, Grace Dinh, Mahesh Subedar, Yakun Sophia Shao<br><i>56th IEEE/ACM International Symposium on Microarchitecture (MICRO), October 2023</i>                                |
| <b>Learning A Continuous and Reconstructible Latent Space for Hardware Accelerator Design</b> ( <a href="#">Paper</a> ) ( <a href="#">Slides</a> ) ( <a href="#">GitHub</a> )<br>Qijing Huang, Charles Hong, John Wawrzynek, Mahesh Subedar, Yakun Sophia Shao<br><i>International Symposium on Performance Analysis of Systems and Software (ISPASS), May 2022</i> |

## PREPRINTS

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| <b>Autocomp: A Powerful and Portable Optimizer for Tensor Accelerators</b> ( <a href="#">Paper</a> ) ( <a href="#">Blog</a> ) ( <a href="#">GitHub</a> )<br>Charles Hong, Sahil Bhatia, Alvin Cheung, Yakun Sophia Shao<br><i>arXiv, May 2025 (under submission)</i> |
| <b>Polaris: Multi-Fidelity Design Space Exploration of Deep Learning Accelerators</b> ( <a href="#">Paper</a> )<br>Chirag Sakhuja, Charles Hong, Calvin Lin<br><i>arXiv, December 2024</i>                                                                           |

## WORKSHOPS AND RESEARCH COMPETITIONS

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| <b>Autocomp: LLM-Driven Code Optimization for Tensor Accelerators</b> ( <a href="#">Paper/Talk/Slides</a> )<br><b>Outstanding Paper Award</b><br>Charles Hong, Sahil Bhatia, Alvin Cheung, Yakun Sophia Shao<br><i>ML for Computer Architecture and Systems Workshop (MLArchSys), co-located with ISCA, June 2025</i> |
| <b>LLM-Aided Compilation for Tensor Accelerators</b> ( <a href="#">Paper</a> )<br>Charles Hong, Sahil Bhatia, Altan Haan, Shengjun Kris Dong, Dima Nikiforov, Alvin Cheung, Yakun Sophia Shao<br><i>The First IEEE International Workshop on LLM-Aided Design (ISLAD), June 2024</i>                                  |

## Sample-Efficient Mapspace Optimization for DNN Accelerators with Bayesian Learning (Paper)

Grace Dinh, Iniyaal Kannan, Hengrui Luo, Charles Hong, Younghyun Cho, James Demmel, Sherry Li, Yang Liu  
ML for Computer Architecture and Systems Workshop (MLArchSys), co-located with ISCA, June 2023

## Modeling DNN Layer Performance Across Accelerators (Paper) (Poster)

Charles Hong, Yakun Sophia Shao  
MICRO ACM Student Research Competition, October 2021

## EXPERIENCE

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### University of California, Berkeley

June 2020 – Present

Undergraduate/Graduate Student Researcher

Berkeley, CA

- Researcher at ADEPT and SLICE computer architecture labs, advised by Professor Sophia Shao.
- Areas of focus include deep learning accelerators, hardware/software co-design, and ML for systems (see publications and projects for details).

### Siemens EDA

May 2024 - August 2024

HLS Intern

Wilsonville, OR

- Improved tool flow for Catapult HLS-generated machine learning acceleration and demonstrated results via FPGA prototype-based demo.

### University of California, Berkeley

August 2018 – August 2023

Undergraduate/Graduate Student Instructor

Berkeley, CA

- Lecturer for CS 61C in Summer 2023.
- Six-time (U)GSI for CS 61A (programming fundamentals), CS 61C (computer architecture), and EECS 151 (digital design, FPGAs).

### Intel Labs

May 2022 - December 2022

AI Research Intern

Remote

- Developed methods for AI/ML-based optimization of deep learning accelerator hardware and software.
- Prototyped an Intel FPGA-based platform for SoC evaluation.

### Apple

May 2021 – August 2021

Hardware Technology Intern

Cupertino, CA

- Intern in CPU DV. Developed CPU Emulation Flow 2.0, which enhances support for internal emulation users, increases test throughput, and reduces runtime of common tasks.

### NVIDIA

May 2020 – August 2020

System Architect Intern

Santa Clara, CA

- Profiled CPU networking workloads and built a machine learning-based cost model, predicting cycle counts to within 15% accuracy.

### NVIDIA

May 2019 – August 2019

System Architect Intern

Santa Clara, CA

- Built tool to model power data of SoC workloads based on hardware configuration and task scheduling across various IPs.
- Enabled estimation of power variation over time and sped up evaluation by several times.

### Oski Technology

June 2018 – August 2018

Formal Verification Intern

San Jose, CA

- Verified an interconnect bridge design and prototyped Chisel FV flow.

## TECHNICAL SKILLS

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**Software Engineering:** Python, Java, C/C++

**Web Development:** HTML/CSS/JavaScript

**Hardware and Systems Engineering:** Expert in computer architecture, especially domain-specific accelerators. Experienced in Verilog/SystemVerilog RTL design. Familiar with compilers, egraphs, code generation.

**Machine Learning/Data Science:** Experienced in model training from sklearn to PyTorch to HuggingFace Transformers. Experienced in data analysis and visualization with NumPy, Pandas, Matplotlib.

**EDA Tools:** FPGAs (Xilinx Vivado and Altera Quartus), Cadence Joules

**Formal Methods:** Familiar with Rosette and Z3 for program synthesis and verification, as well as JasperGold for RTL FV.

## VOLUNTEERING

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### Pioneers in Engineering

September 2017 – Present

*Advisor, Director of Engineering, Project Manager, Software Developer*

*Berkeley, CA*

- Various roles helping run robotics competition for local underserved high schools.
- Two years as project manager; one year as engineering director, managing 30+.
- Developed robot simulator web app ([pimulator.pierobotics.org](http://pimulator.pierobotics.org)).

## OTHER PROJECTS

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### Rewrite Rule Synthesis: A Survey ([Paper/Slides](#))

Charles Hong

*CS 294-260 (Declarative Program Analysis and Optimization) Class Project, Spring 2024*

### Transformers in Reinforcement Learning for Hardware Design Space Exploration ([Paper](#))

Junsun Choi, Charles Hong

*CS 285 (Deep Reinforcement Learning) Class Project, Fall 2022*

### Fast Thread Migration in a Heterogeneous ISA System ([Paper](#)) ([Poster](#))

Max Banister, John Fang, Charles Hong

*CS 262A (Advanced Computer Systems) Class Project, Fall 2021*

### Enhancing Yelp Rating Predictions ([Paper](#))

Charles Hong, Kevin Zhu

*CS 182 (Deep Learning) Class Project, Spring 2021*

### A Chipyard Comparison of NVDLA and Gemmini ([Paper](#))

Abraham Gonzalez, Charles Hong

*EE 290-2 (Hardware for Machine Learning) Class Project, Spring 2020*